BISHWASH KHANAL

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PROFILE

AI Engineer (Machine Learning Engineer) with five years of experience building machine learning systems and taking them to production. I work on fine-tuning and compression of open-weight large language models (LLMs), retrieval-augmented generation (RAG) and multi-agent applications, real-time computer vision on edge hardware, and AI governance for AI-native software engineering. I hold an M.Sc. in Artificial Intelligence from the University of Jyväskylä (2026) and have five publications, two of them peer-reviewed, in computer vision, efficient LLMs, and multi-agent code generation. Based in Espoo, Finland.

WORK EXPERIENCE

Project Researcher - ANSE Project Jan 2026 - Present
University of Jyväskylä (full-time, hybrid) Jyväskylä, Finland
Research Assistant Jan - Jun 2026; promoted to Project Researcher Jul 2026

- Build and evaluate multi-agent systems for code generation under test-driven development, orchestrating LLM agents with LangGraph and gating each step on automated Pytest runs
- Design and iterate on TDD orchestration workflows for AI-native software engineering
- Ensure AI-generated software artifacts comply with AI governance requirements, embedding traceability, automated verification, and human-oversight checkpoints into the agentic development workflow, aligning with EU AI Act and ISO/IEC 42001 requirements
- Researched failure modes in agentic TDD pipelines and proposed practical mitigations; co-authored *TDD Governance for Multi-Agent Code Generation via Prompt Engineering*, published at Prompt SE @ EASE 2026
- **Technologies:** Python, LangGraph, OpenAI API, Pytest, Docker

Artificial Intelligence Engineer Mar 2024 - Aug 2025
OptiML (founding team member; part-time, remote) California, USA

- Developed custom pipelines for fine-tuning and compressing open-weight LLMs (Llama, Qwen, Mistral, Gemma, Phi, DeepSeek), applying pruning methods (SparseGPT, Wanda, magnitude pruning) at up to 50% sparsity and showing that task-specific calibration data significantly improves downstream performance of compressed models
- Researched and evaluated state-of-the-art LLM compression techniques and published the results as *Evaluating the Impact of Compression Techniques on Task-Specific Performance of Large Language Models* (arXiv:2409.11233)
- Integrated model compression into the fine-tuning workflow to support efficient deployment of large models at scale
- Set up the training stack, experiment tracking, and evaluation harness as founding engineer
- **Technologies:** PyTorch, Transformers, PEFT, BitsAndBytes, Accelerate, Weights & Biases

Computer Vision Engineer Mar 2021 - Aug 2024
E.K. Solutions Pvt. Ltd. (full-time, on-site) Lalitpur, Nepal

- Quantized and deployed face detection, recognition, and swapping models (YOLOv8, ArcFace, Inswapper, StyleGAN2) on Jetson AGX Orin using INT8/FP16 quantization with PyTorch, ONNX Runtime, and TensorRT, reaching real-time performance of up to 36 FPS ($\sim 2\times$ over the FP32 CUDA baseline) at 640×480 through precision tuning, execution-provider selection, and batched multi-face post-processing
- Developed an early-adoption RAG chatbot integrating GPT-4 for project management, database analysis, and bulk CV screening using LangChain, ChromaDB, and FastAPI, deployed internally across a 300-employee organization
- Generated planar simplified 3D meshes for virtual tours from iPad Pro LiDAR captures, implementing custom algorithms for planar simplification, ICP registration, TSDF volumetric integration, and MVS texturing
- Applied RandLA-Net for semantic segmentation and developed the full-stack pipeline using Python, C++, Swift, and Three.js
- **Technologies:** PyTorch, TensorRT, ONNX, OpenCV, Open3D, LangChain, FastAPI, Celery, NumPy

Freelance & Contract Work 2023
Computer vision automation for GoThru Media (automated pose estimation between uncalibrated spherical panoramas, removing a manual production step); LLM training-data evaluation for Scale AI.

SKILLS

Programming: Python, C++, C, SQL, Java, MATLAB
LLMs & Generative AI: Fine-tuning (LoRA/PEFT), quantization and model compression, retrieval-augmented generation (RAG), agentic and multi-agent systems, prompt engineering, LLM evaluation, AI governance & responsible AI; Hugging Face Transformers, LangChain, LangGraph, OpenAI API, vLLM, MCP (Model Context Protocol)
ML & Computer Vision: PyTorch, TensorFlow, CNNs, GANs, Transformers, OpenCV, Open3D, TensorRT, ONNX
Cloud & MLOps: Azure / Azure ML, AWS, GCP / Vertex AI, Docker, Kubernetes, Git, CI/CD (GitHub Actions), automated

testing (Pytest), Weights & Biases, FastAPI
Data: PostgreSQL, MySQL, ChromaDB, vector search, NumPy, Pandas

PROJECTS

- **Shared-Expert LoRA Adapters for Domain-Specific Photographic Assessment** Oct 2025 - May 2026
  Designed a modular parameter-efficient architecture (always-active rank-64 shared LoRA expert + query-routed rank-32 specialists on frozen Qwen3-VL-8B) that turns a general vision–language model into a domain-expert photographic critic, nearly doubling BERTScore-F1 over monolithic LoRA (0.23 $\rightarrow$ 0.42) and winning 84.6% of GPT-4o-judged pairwise comparisons with fewer active parameters. M.Sc. thesis work, released as arXiv:2608.17514 with code and project page.
- **Lawgic: Intelligent Legal Document Analysis Platform** Aug 2025 - Jan 2026
  Developed a full-stack legal document analysis platform using FastAPI and React, integrating an external OCR microservice for processing PDF uploads. Developed a conversational AI interface with features for multi-lingual document translation, semantic annotations, and automated legal analysis.
- **Multi-Agent System for Automated Literature Review** Mar 2025 - May 2025
  Built a multi-agent system that turns a natural-language research question into a synthesized literature report: agents generate arXiv queries, filter results for relevance, retrieve and store papers, extract methodology and findings, and synthesize common themes and research gaps. Implemented with BDI-style agent reasoning (SPADE, AgentSpeak), the Gemini API, and the arXiv and Jina Reader APIs.

EDUCATION

M.Sc. in Artificial Intelligence Aug 2024 - Jul 2026
University of Jyväskylä, Finland
Grade: 5/5 ; JYU Rector's Scholarship

B.Sc. in Electrical and Computer Engineering Sep 2017 - Aug 2020
Jacobs University Bremen, Germany
Grade: 1.73; Merit-Based Scholarship for academic achievement; Minor: Intelligent Mobile Systems

PUBLICATIONS

- **[Preprint]** B. Khanal, A. Zhang, S. Tarkoma, T. Mikkonen, A. Kumar (2026). SE-MoLoRA: Shared-Expert LoRA Adapters for Domain-Specific Photographic Assessment. *arXiv:2608.17514.*
- **[Peer-reviewed journal]** B. Khanal, M. Om, S. Rijal, V. Ojha (2024). Alignment of a 360° image with posed color images for locally accurate texturing of 3D mesh. *Frontiers in Computer Science*, Vol. 6, ISSN: 2624-9898.
- **[Peer-reviewed conference]** T. Hasanli, S. Siddeeq, B. Khanal, P. Kotilainen, T. Mikkonen, P. Abrahamsson (2026). TDD Governance for Multi-Agent Code Generation via Prompt Engineering. *Prompt SE @ EASE 2026* (International Conference on Evaluation and Assessment in Software Engineering), June 2026.
- **[Preprint]** B. Khanal, S. Rijal, M. Awale, V. Ojha (2024). Structure-preserving Planar Simplification for Indoor Environments. *arXiv:2408.06814.*
- **[Preprint]** B. Khanal, J. M. Capone (2024). Evaluating the Impact of Compression Techniques on Task-Specific Performance of Large Language Models. *arXiv:2409.11233.*

LANGUAGES

English (fluent), Nepali (native), German (A2), Finnish (basics)